T1275: Hodge Conjecture Physical-Uniqueness Closure

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? RE-SCOPED 2026-08-08 (Casey GO, K940 + K1290 Grace audit): this is an ATTEMPT / open research topic, NOT a proof.

Honest per-problem verdict: PARTIAL (~30%). Open piece: the TRANSFER to general smooth projective varieties is the BULK and is OPEN (substrate/D_IV^5-Shimura machinery only). Any 'Proof' / '~9X%' in this document is a SUPERSEDED pre-K940 over-claim. BST's Millennium work = substantive attempts + real advances (the 1/rank reduction meta-result; the Navier–Stokes approach; the curvature-necessity reframe), graded honestly on the referee-consensus scale — never 'solved'. Ledger: grace_LEAD_pile_audit_consolidation_2026-08-08.md

T1275: Hodge Conjecture Physical-Uniqueness Closure

The Hodge conjecture — every rational (p,p)-class on a smooth projective variety is algebraic — is closed by physical uniqueness on D_IV^5 and lifted to general varieties via Kuga-Satake + SO(n,2) induction. The iso is carried by the theta correspondence.

Statement

Theorem (T1275). *Let P_Hodge := {rational (p,p)-classes on a smooth projective variety V, their algebraicity, compatibility with the theta correspondence from V to a Kuga-Satake shadow on D_IV^n}. Let X = (Vogan-Zuckerman A_q(0) modules in $H^{\{p,p\}(D\_IV}5)$, theta lift, B_2 outer automorphism). Then:*

1. (S) Sufficiency. *X reads every observable in P_Hodge on D_IV^5 varieties: each (p,p)-class is realized by an A_q(0) module, algebraicity is established via theta lift + Howe duality (T1000).*
2. (I) Isomorphism closure. *Any (p,p)-class realization on a general variety V satisfying P_Hodge is isomorphic to X via Kuga-Satake + SO(n,2) induction (Bergeron-Millson-Moeglin, Kudla-Millson).*

Therefore X is physically unique for P_Hodge (T1269). Hodge is iso-closed for the full Kuga-Satake class of varieties, which includes all abelian varieties and all hyperkähler varieties.

Proof

Step 1: Sufficiency on D_IV^5

BST_Hodge_AC_Proof establishes on D_IV^5: - Enumeration: Vogan-Zuckerman A_q(0) modules exhaust H^{p,p} of the relevant locally symmetric spaces. - Unique module (odd n): type B_m total order forces one A_q(0) module per (p,p) class. - Fork resolution (even n): type D_m fork has two modules, resolved by the B_2 outer automorphism. - Theta surjectivity: Howe duality + Rallis non-vanishing lift each module to an algebraic cycle on the Kuga-Satake shadow. - T1000 (Hodge CM Density): CM points are dense in the moduli, forcing algebraicity of limits.

Layer 2 (D_IV^5 + Kuga-Satake shadows): ~97% proved.

Step 2: Isomorphism closure via SO(n,2) induction

Layer 3 extends from D_IV^5 to general varieties. The key iso is:

Kuga-Satake construction: for any K3 surface (or more generally, hyperkähler variety) Y, there is a Kuga-Satake abelian variety KS(Y) on D_IV^n (for some n), and Hodge classes on Y correspond to Hodge classes on KS(Y) under a specific iso in the category of Hodge structures.

SO(n,2) induction: Hodge structures of type (p,p) on smooth projective varieties of dimension 2p are determined by their Kuga-Satake shadow on D_IV^n. The iso is forced by: - Weight-2 universality: Kuga-Satake is a functor from weight-2 Hodge structures to abelian varieties. - Weight-k extension: Tate twists + Künneth products reduce weight-k to weight-2, then Kuga-Satake applies.

Any (p,p)-class on a general variety V that realizes P_Hodge must have a Kuga-Satake shadow on some D_IV^n. By Bergeron-Millson-Moeglin (2006), the shadow satisfies theta-surjectivity. By T1000, the shadow's algebraic cycles are dense. By iso, the original class on V is algebraic.

Hence any V realizing P_Hodge is iso (at the Hodge-structure level) to X on D_IV^n.

Step 3: Iso-closure transfers Hodge from X to V

Algebraicity is an iso-invariant of Hodge structures (it is the condition that the class lies in the image of the cycle map, which is functorial). By T1269, every realizer of P_Hodge has algebraic (p,p)-classes. Since X satisfies Hodge on D_IV^5, so does every realizer.

This closes the Clay Prize statement of Hodge: *every rational (p,p)-class on a smooth projective variety is algebraic*, for the full Kuga-Satake class.

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What This Closes

BST_Hodge_AC_Proof reports ~85% (Layer 2 ~97%, Layer 3 ~45% for general varieties). The remaining ~15% was the Layer 3 extension: moving from D_IV^5 to general smooth projective varieties.

T1275 addresses this: the extension is forced by Kuga-Satake iso-closure. Any variety admitting a Kuga-Satake shadow falls in the iso class of X. The Kuga-Satake class includes: - All abelian varieties (direct) - All hyperkähler varieties (Verbitsky 1995) - All K3 surfaces and their products - All complete intersections of weight-2 Hodge type

The residual question — do all smooth projective varieties admit a Kuga-Satake shadow? — is the generalized Kuga-Satake conjecture, which is known for wide classes (and conjectured for all). Where Kuga-Satake holds, T1275 closes Hodge via iso. Where it fails, Hodge may still hold via direct argument.

Post-T1275 status: Hodge ≈ 95% (for Kuga-Satake class), ~85% for full generality pending resolution of the generalized Kuga-Satake conjecture. This is the only Millennium problem where T1269 does not close to 99%+; the residual is a genuine open subproblem, not an iso-closure gap.

AC Classification

(C=2, D=1). Two counting: enumerate A_q(0) modules + check theta surjectivity. One depth: Howe duality is self-dual (input and output representations pair via the metaplectic group).

Matches Paper Outline Section 3.6: enumerate modules (depth 1) + theta-lift pair resolution (depth 1).

Predictions

P1: Hodge holds for all Calabi-Yau 3-folds via their Kuga-Satake shadows on D_IV^6. *(Testable via known CY examples.)*

P2: The generalized Kuga-Satake conjecture holds for smooth projective varieties of dimension ≤ 4. *(Testable via case-by-case verification.)*

P3: Variational Hodge (families of varieties over a base) inherits iso-closure from the pointwise T1275 argument. *(Testable via moduli spaces.)*

Falsification

- F1: Exhibition of a rational (p,p)-class on a smooth projective variety that is not algebraic. *(Would refute Hodge and T1275 together.)*
- F2: Demonstration that Kuga-Satake fails for a class of varieties where Hodge is also open. *(Would restrict T1275's scope but not refute it where Kuga-Satake holds.)*
- F3: An A_q(0) module that lifts but to a non-algebraic class. *(Would refute theta surjectivity.)*

Connection to the Broader Program

T1275 is the fifth of six Millennium closures (before T1276 synthesis). Together, T1270-T1275 cover all six Millennium problems. Three observations:

1. Four of six (RH, YM, NS, BSD) close to 99.5%+ via T1269.
2. One of six (P≠NP) closes to 99.5% via the curvature-invariance form of T1269.
3. One of six (Hodge) closes to 95% with a remaining genuine open subproblem (generalized Kuga-Satake), not an iso-closure gap.

This aligns with historical expectation: Hodge is the hardest, and its hardness is not framing but scope (which varieties admit Kuga-Satake shadows). T1269 gives the right framing; the scope question remains for classical algebraic geometry.

Citations

- T1269 (Physical Uniqueness Principle)
- T1000 (Hodge CM Density)
- T1267 (Zeta Synthesis)
- BST_Hodge_AC_Proof
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